Probeklausur Statistik mit R - Comprehensive Exam

- This mock exam is meant to take 90 minutes in total
- You are allowed to use a cheat sheet (one A4 page, both sides)
- These questions cover all topics from the course syllabus
- Show all calculations where required
- For multiple choice questions, mark only ONE answer

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Aufgabe 1: Multiple Choice Questions (30 points, 2 points each)

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Only one of the given options is correct. If you need to correct your choice, put the word "correct" next to the box.

1. A variable measured on a nominal scale...
   - ☐ allows for meaningful arithmetic operations like addition and subtraction
   - ☐ has equal intervals between values but no true zero point
   - ☐ represents categories without any inherent order or ranking
   - ☐ can be used to calculate means and standard deviations
   - ☐ must have at least 5 categories to be valid

2. The Central Limit Theorem states that...
   - ☐ the population distribution is always normal
   - ☐ sampling distributions of means approach normality as sample size increases
   - ☐ individual observations in a sample are normally distributed
   - ☐ variance decreases as sample size increases
   - ☐ the mean of the population equals the mean of the sample

3. Type I error is...
   - ☐ failing to reject the null hypothesis when it is false
   - ☐ rejecting the null hypothesis when it is true
   - ☐ correctly rejecting the null hypothesis
   - ☐ correctly accepting the null hypothesis
   - ☐ controlled by beta (β)

4. A 95% confidence interval for a mean...
   - ☐ contains the true population mean with 95% probability
   - ☐ will contain the true mean in 95 out of 100 repeated experiments
   - ☐ contains 95% of all observations in the sample
   - ☐ has a width that is independent of sample size
   - ☐ proves that the null hypothesis is false

5. When should you use Fisher's Exact Test instead of Chi-square?
   - ☐ when you have more than 2 categorical variables
   - ☐ when expected frequencies in cells are less than 5
   - ☐ when you have ordinal data
   - ☐ when the sample size is greater than 100
   - ☐ when testing for normality

6. Cohen's d measures...
   - ☐ the correlation between two variables
   - ☐ the standardized difference between two means
   - ☐ the probability of Type II error
   - ☐ the effect of sample size on power
   - ☐ the homogeneity of variance

7. In regression, $R^2$ represents...
   - ☐ the correlation coefficient squared
   - ☐ the proportion of variance in Y explained by the model
   - ☐ the residual sum of squares
   - ☐ the standard error of the regression
   - ☐ the strength of the relationship between X and Y only

8. In a mixed-effects model, random effects...
   - ☐ are the same as fixed effects but estimated differently
   - ☐ account for variability across grouping units (e.g., subjects, items)
   - ☐ should only be used when you have more than 1000 observations
   - ☐ are always uncorrelated with fixed effects
   - ☐ must include both intercepts and slopes

9. In Bayesian statistics, the posterior distribution...
   - ☐ is the same as the likelihood function
   - ☐ represents our beliefs before seeing the data
   - ☐ is calculated as Prior × Likelihood / Evidence
   - ☐ is always a normal distribution
   - ☐ does not depend on the prior

10. A Bayes Factor ($BF_{10}$) of 10 means...
    - ☐ the null hypothesis is 10 times more likely than the alternative
    - ☐ the alternative hypothesis is 10 times more likely than the null
    - ☐ the posterior probability is 0.10
    - ☐ there is a 10% chance the null hypothesis is true
    - ☐ you need 10 more observations to reach a conclusion

11. McNemar's Test is used when...
    - ☐ comparing two independent proportions
    - ☐ comparing paired categorical data (e.g., before/after)
    - ☐ testing for correlation between continuous variables
    - ☐ you have more than two categorical variables
    - ☐ expected frequencies are less than 5

12. Which assumption is NOT required for a standard t-test?
    - ☐ normality of the data
    - ☐ independence of observations
    - ☐ equal sample sizes in both groups
    - ☐ continuous dependent variable

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    □ homogeneity of variance (for independent samples)

13. In a factorial ANOVA, an interaction means...
    □ the two factors are correlated
    □ the effect of one factor depends on the level of another factor
    □ both main effects must be significant
    □ the data violate the assumption of independence
    □ you need to use a different statistical test

14. Random slopes in a mixed model...
    □ allow the effect of a predictor to vary across groups
    □ are required whenever you have random intercepts
    □ make the model easier to converge
    □ should only be used for between-subjects factors
    □ increase Type I error rates

15. Poisson regression is appropriate when...
    □ the dependent variable is continuous and normally distributed
    □ the dependent variable consists of count data
    □ you have binary outcomes (0 or 1)
    □ the residuals are heteroscedastic
    □ you want to test for normality

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Aufgabe 2: Distributions and Hypothesis Testing (15 points)
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Below you see two graphs for a t-distribution with 20 degrees of freedom.

[Imagine: Left graph shows density function, Right graph shows cumulative distribution]

1. (3 points) Explain the difference between what the density function shows
   versus what the cumulative distribution function shows.

2. (4 points) For a one-tailed test with α = 0.05 (considering only positive
   t values), draw the rejection region in the density plot and explain how
   you determined its location.

3. (4 points) For a two-tailed test with α = 0.05, draw both rejection regions
   and explain.

4. (2 points) You obtain t = 1.85 with df = 20. Using the table provided, what
   is your conclusion for a two-tailed test at α = 0.05?
   [Critical value for two-tailed test, df=20, α=.05 is 2.086]

5. (2 points) What would be your conclusion for a one-tailed test (positive
   direction) with the same t-value?
   [Critical value for one-tailed test, df=20, α=.05 is 1.725]
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Aufgabe 3: Test Selection and Justification (20 points)
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For each scenario, indicate which statistical test is most appropriate and
provide TWO specific reasons why this test is suitable (or why alternatives
are not). Consider aspects like: data type, independence, sample size,
assumptions, research question.

Scenario A (5 points):
A researcher wants to know if there is a relationship between gender (male/
female) and smartphone preference (iPhone/Android/Other). Data were collected
from 150 randomly selected individuals.

Test: _________________

Reason 1:
Reason 2:

Scenario B (5 points):
A psychologist measures reaction time (in milliseconds) for 25 participants
before and after drinking coffee. She wants to test if coffee affects reaction
time.

Test: _________________

Reason 1:
Reason 2:

Scenario C (5 points):
A linguist recorded reading times (in seconds) for 30 participants reading 50
different sentences. Each participant read all 50 sentences. The research
question is whether sentence length (short vs. long) affects reading time,
accounting for individual differences in reading speed and differences in
sentence difficulty.

Test: _________________

Reason 1:
Reason 2:

Scenario D (5 points):
A medical researcher wants to compare the number of hospital visits (count
data: 0, 1, 2, 3, ...) across three different treatment groups (n=40 per
group). The data show that variance is much larger than the mean.

Test: _________________
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Reason 1:
Reason 2:

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Aufgabe 4: Bayesian Statistics (15 points)
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You are conducting an experiment where participants taste-test a new chocolate recipe. You want to know the probability that people prefer the new recipe over the old one. You collect data from 50 participants: 32 prefer the new recipe, 18 prefer the old recipe.

You have three possible priors for the probability p (that someone prefers the new recipe):

Prior A: Uniform distribution [0, 1] - flat prior
Prior B: Beta(5, 5) - peaked around 0.5
Prior C: Beta(8, 2) - peaked around 0.8

1. (6 points) For each prior, classify it as non-informative, weakly informative, or strongly informative, and explain your classification in the context of this specific problem:

   Prior A:

   Prior B:

   Prior C:

2. (3 points) You run a Bayesian analysis and obtain a 95% credible interval of [0.51, 0.77] for p. Interpret this interval. What does it mean in plain language?

3. (3 points) Your colleague uses a frequentist approach and reports: "We reject the null hypothesis that p = 0.5, p < 0.05". Explain how the Bayesian credible interval and the frequentist p-value differ in their interpretation.

4. (3 points) What diagnostic checks would you perform to ensure your Bayesian MCMC sampling was successful? Name two specific diagnostics and their criteria.

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Aufgabe 5: Linear Mixed-Effects Model (LMER) (30 points)
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A researcher investigated whether background music affects concentration on a

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memory task. The study design:

- 40 participants (20 young adults aged 20-30, 20 older adults aged 60-70)
- Each participant completed the memory task under 3 conditions:
  * No music (silence)
  * Classical music
  * Pop music
- Each condition was tested on 4 different days (to account for day-to-day
  variation)
- Dependent variable: Number of words recalled (out of 20)

Variables:
- recalled: number of words recalled (0-20)
- music_condition: Silence (reference), Classical, Pop
- age_group: Young (reference), Old
- participant: 40 participants
- day: 4 testing days per condition per participant

The researcher fitted the following model:

model <- lmer(recalled ~ music_condition * age_group +
              (music_condition | participant) +
              (1 | day),
              data = memory_data)

summary(model)

Linear mixed model fit by REML ['lmerMod']
Formula: recalled ~ music_condition * age_group + (music_condition | participant) + (1 | day)
   Data: memory_data

Random effects:
 Groups      Name                    Variance Std.Dev. Corr
 participant (Intercept)             4.250    2.062
             music_conditionClassical 0.680    0.825   -0.34
             music_conditionPop       1.120    1.058   -0.28  0.52
 day         (Intercept)             0.450    0.671
 Residual                            2.180    1.476
Number of obs: 480, groups:  participant, 40; day, 4

Fixed effects:
                                  Estimate Std.Error df     t value Pr(>|t|)
(Intercept)                        14.250    0.425   40.2   33.53  <0.001 ***
music_conditionClassical            0.850    0.320   38.5    2.66   0.011 *
music_conditionPop                 -1.450    0.360   38.8   -4.03  <0.001 ***
age_groupOld                       -2.150    0.600   38.0   -3.58   0.001 **
music_conditionClassical:age_groupOld  -0.420  0.450  152.3  -0.93   0.352
music_conditionPop:age_groupOld     1.280    0.510   154.1   2.51   0.013 *
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Correlation of Fixed Effects:
              (Intr) ms_cC ms_cP ag_grO ms_cC:_O
msc_cndtnCl   -0.376
msc_cndtnPp   -0.336  0.520
age_groupOld  -0.707  0.266  0.238
ms_cndtnC:_O   0.267 -0.707 -0.369 -0.376
ms_cndtnP:_O   0.236 -0.366 -0.707 -0.336  0.520

Questions:

1. (4 points) Identify and explain all random effects in this model:
   a) What random intercepts are included and what do they represent?
   b) What random slopes are included and what do they represent?

2. (6 points) Interpret the following random effects estimates:
   a) The participant (Intercept) variance is 4.250. What does this tell us?
   b) The correlation between (Intercept) and music_conditionPop is -0.28.
      Interpret this correlation.
   c) Why is there only a random intercept for day, not random slopes?

3. (8 points) Using the fixed effects estimates, calculate the predicted number
   of words recalled for the following (assume random effects = 0):

   a) A young adult in silence (show calculation)

   b) A young adult listening to classical music (show calculation)

   c) An older adult listening to pop music (show calculation)

4. (4 points) Interpret the significant interaction effect
   music_conditionPop:age_groupOld (β = 1.280, p = 0.013). What does this
   tell us about how pop music affects young vs. old adults?

5. (4 points) The researcher initially tried this maximal model:

   model_max <- lmer(recalled ~ music_condition * age_group +
                     (music_condition * age_group | participant) +
                     (music_condition * age_group | day),
                     data = memory_data)

   But received a "singular fit" warning.
   a) Why is the original model (shown above) more appropriate than this
      maximal model?
   b) Name one specific reason the maximal model failed.

6. (4 points) Based on this design:
   a) Is music_condition a within-subjects or between-subjects factor? Explain.
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   b) Is age_group a within-subjects or between-subjects factor? Explain.

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Aufgabe 6: Regression Diagnostics and Assumptions (15 points)
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A researcher runs a multiple regression predicting salary (in thousands) from
years of experience and education level:

model <- lm(salary ~ experience + education, data = salary_data)

Below are four diagnostic plots from this regression:

[Imagine 4 plots:]
Plot A: Residuals vs Fitted - shows a clear funnel shape (increasing spread)
Plot B: Q-Q plot - points deviate from the line at both tails
Plot C: Scale-Location plot - shows increasing trend (red line goes up)
Plot D: Residuals vs Leverage - one point has Cook's distance > 0.5

1. (8 points) For each plot, identify which assumption is being tested and
   whether it appears to be violated. If violated, suggest a remedy:

   Plot A:
   Assumption:
   Violated?
   Remedy:

   Plot B:
   Assumption:
   Violated?
   Remedy:

   Plot C:
   Assumption:
   Violated?
   Remedy:

   Plot D:
   Issue identified:
   What should you do?

2. (4 points) The researcher reports VIF values:
   - experience: VIF = 8.2
   - education: VIF = 7.9

   a) What problem do these VIF values suggest?
   b) Why is this problematic for the regression?

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3. (3 points) After addressing the violations, the researcher compares two models:

   Model 1: salary ~ experience + education
   Model 2: salary ~ experience + education + experience:education

   She uses anova(model1, model2) and gets F(1,96) = 8.45, p = 0.005.

   a) What does this test tell us?
   b) Which model should she choose?

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Aufgabe 7: Calculations - Paired t-test (20 points)
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A cognitive psychologist wants to test whether a memory training program improves recall. She tests 10 participants before and after the training:

| Participant | Before | After | Difference (After - Before) |
|---|---|---|---|
| 1 | 12 | 15 | 3 |
| 2 | 14 | 16 | 2 |
| 3 | 10 | 14 | 4 |
| 4 | 13 | 15 | 2 |
| 5 | 11 | 13 | 2 |
| 6 | 15 | 18 | 3 |
| 7 | 9 | 13 | 4 |
| 8 | 14 | 17 | 3 |
| 9 | 12 | 14 | 2 |
| 10 | 13 | 16 | 3 |

1. (6 points) Calculate the mean difference and standard deviation of differences. Show all steps:

   Mean difference (M_d) =

   Standard deviation (SD_d) =

2. (4 points) Calculate the t-statistic for testing $H_0$: μ_d = 0

   Formula: t = M_d / (SD_d / √n)

   Calculation:

3. (3 points) What are the degrees of freedom for this test?

4. (3 points) Using the critical value from the table (t_critical = 2.262 for df=9, α=.05, two-tailed), what is your conclusion? State your decision

about $H_0$ and interpret the result in context.

5. (4 points) Calculate Cohen's d for this paired sample:

   Formula: d = M_d / SD_d

   Calculation:

   Interpretation (small/medium/large):

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Aufgabe 8: Categorical Predictors and Coding (15 points)
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A researcher studies the effect of teaching method on exam scores. Three methods are compared: Traditional (T), Flipped Classroom (F), and Online (O).

She uses this model in R:
```
model <- lm(exam_score ~ teaching_method, data = teaching_data)
```

With treatment (dummy) coding (Traditional = reference category):

Coefficients:

| | Estimate | Std. Error | t value | Pr(>\|t\|) |
|---|---|---|---|---|
| (Intercept) | 72.40 | 1.85 | 39.14 | <0.001 *** |
| teaching_methodFlipped | 8.60 | 2.52 | 3.41 | 0.001 ** |
| teaching_methodOnline | 3.20 | 2.48 | 1.29 | 0.201 |

1. (6 points) Interpret each coefficient:

   a) (Intercept) = 72.40 means:

   b) teaching_methodFlipped = 8.60 means:

   c) teaching_methodOnline = 3.20 means:

2. (3 points) Calculate the predicted exam score for:
   a) A student in the Flipped Classroom:
   b) A student in the Online condition:

3. (3 points) The researcher wants to test if Flipped Classroom is significantly better than Online. This comparison is not directly available in the output above. What should she do? Name the specific statistical procedure.

4. (3 points) If the researcher used sum coding instead of treatment coding, how would the interpretation of the (Intercept) change?

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Aufgabe 9: Generalized Linear Model - Logistic Regression (15 points)
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A researcher studies whether study time (hours) and previous GPA predict
passing an exam (Pass=1, Fail=0). She uses logistic regression:

model <- glm(pass ~ study_hours + gpa, data = exam_data, family = binomial)

summary(model)

Coefficients:
              Estimate  Std. Error  z value  Pr(>|z|)
(Intercept)   -4.200     1.050      -4.00    <0.001 ***
study_hours    0.350     0.085       4.12    <0.001 ***
gpa            1.800     0.520       3.46     0.001 **

1. (4 points) The coefficient for study_hours is 0.350.
   a) What does this mean on the log-odds scale?
   b) Convert this to an odds ratio: OR = exp(0.350) = ____
   c) Interpret the odds ratio in plain language.

2. (5 points) Calculate the predicted probability of passing for a student
   with 5 study hours and GPA = 3.0:

   Step 1: Calculate log-odds (logit):
   logit =

   Step 2: Convert to probability using p = exp(logit) / (1 + exp(logit)):
   p =

3. (3 points) Why is logistic regression appropriate for this problem instead
   of linear regression? Give two specific reasons.

4. (3 points) The residual deviance is 145.2 on 150 df. The null deviance
   (intercept only) is 198.5 on 152 df. Calculate the reduction in deviance
   and explain what this tells us about model fit.

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Aufgabe 10: Short Answer Questions (20 points, 4 points each)
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1. Explain the difference between the Wilcoxon signed-rank test and the
   Mann-Whitney U test. When would you use each?

2. What is the difference between fixed effects and random effects in a
   mixed-effects model? Provide an example of each.

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3. In regression diagnostics, what is Cook's distance and what does it measure? What threshold is typically used to identify problematic observations?

4. Explain what "crossed random effects" means in a mixed model. Give an example of a study design that would require crossed random effects.

5. What is the difference between AIC and BIC for model comparison? Which is more conservative (penalizes complexity more)?

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APPENDIX: Statistical Tables

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t-distribution critical values (two-tailed):

| df | α=.10 | α=.05 | α=.01 |
|---|---|---|---|
| 9 | 1.833 | 2.262 | 3.250 |
| 10 | 1.812 | 2.228 | 3.169 |
| 15 | 1.753 | 2.131 | 2.947 |
| 20 | 1.725 | 2.086 | 2.845 |
| 30 | 1.697 | 2.042 | 2.750 |
| ∞ | 1.645 | 1.960 | 2.576 |

t-distribution critical values (one-tailed):

| df | α=.10 | α=.05 | α=.01 |
|---|---|---|---|
| 9 | 1.383 | 1.833 | 2.821 |
| 10 | 1.372 | 1.812 | 2.764 |
| 15 | 1.341 | 1.753 | 2.602 |
| 20 | 1.325 | 1.725 | 2.528 |
| 30 | 1.310 | 1.697 | 2.457 |
| ∞ | 1.282 | 1.645 | 2.326 |

Chi-square critical values:

| df | α=.05 | α=.01 |
|---|---|---|
| 1 | 3.841 | 6.635 |
| 2 | 5.991 | 9.210 |
| 3 | 7.815 | 11.345 |
| 4 | 9.488 | 13.277 |
| 5 | 11.070 | 15.086 |

F-distribution critical values (α=.05):

| | df2 | | | | |
|---|---|---|---|---|---|
| df1 | 10 | 20 | 30 | 50 | 100 |
| 1 | 4.96 | 4.35 | 4.17 | 4.03 | 3.94 |
| 2 | 4.10 | 3.49 | 3.32 | 3.18 | 3.09 |
| 3 | 3.71 | 3.10 | 2.92 | 2.79 | 2.70 |
| 4 | 3.48 | 2.87 | 2.69 | 2.56 | 2.46 |

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END OF EXAM

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Total Points: 195 points

Time: 90 minutes

Remember to:

- Show all calculations
- Explain your reasoning
- Use the statistical tables provided
- Check your work

Good luck!